

MORA, MN, USA

WMO: 727475

Lat: 45.886N

Lon: 93.272W

Elev: 1012

StdP: 14.17

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04940

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.9	-11.4	-26.3	1.3	-17.4	-19.6	1.9	-10.3	21.9	13.0	19.5	15.7	2.3	320	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.2	87.8	72.6	84.4	70.0	81.3	67.7	74.6	83.5	72.4	81.2	70.3	77.8	9.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.0	123.1	81.1	69.9	114.2	78.2	67.7	105.7	75.5	38.8	83.3	36.6	80.6	34.8	77.1	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.3	16.9	14.8	-24.1	91.9	6.7	3.4	-29.0	94.3	-32.9	96.3	-36.7	98.2	-41.6	100.7		
(5)			-23.9	78.1	6.6	2.4	-28.7	79.8	-32.6	81.2	-36.3	82.6	-41.1	84.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.5	12.5	16.6	28.9	42.3	54.2	63.9	68.7	66.1	58.4	45.5	32.8	18.8
	DBStd	21.63	13.35	12.86	11.90	9.18	8.06	6.75	5.70	5.69	8.28	8.31	9.43	11.83
	HDD50	4762	1163	936	659	261	46	1	0	0	19	192	518	967
	HDD65	8547	1628	1356	1120	681	349	98	26	55	230	607	965	1432
	CDD50	2029	0	0	4	30	175	417	580	499	270	51	3	0
	CDD65	341	0	0	0	1	13	64	141	89	31	2	0	0
	CDH74	3388	0	0	0	16	217	699	1345	793	298	20	0	0
	CDH80	915	0	0	0	3	57	193	396	194	71	1	0	0
(14) Wind	WSAvg	5.4	5.7	5.7	5.8	6.3	6.0	5.0	4.5	4.1	4.9	5.8	5.8	5.5
(15) Precipitation	PrecAvg	28.4	0.9	0.7	1.5	2.4	3.5	4.1	3.8	3.7	3.1	2.4	1.7	0.9
	PrecMax	41.4	2.9	2.2	4.3	5.7	7.7	8.9	9.0	13.4	7.7	8.5	6.2	3.0
	PrecMin	14.6	0.1	0.0	0.2	0.1	0.9	0.9	0.9	0.7	0.4	0.2	0.0	0.1
	PrecStd	6.5	0.7	0.6	0.9	1.3	1.6	1.9	1.9	2.0	1.9	2.0	1.3	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.5	48.4	64.4	75.2	86.2	90.7	91.2	90.0	86.4	76.8	63.9	45.6
		MCWB	38.3	42.6	52.5	57.0	65.9	73.9	75.7	72.6	69.7	61.6	53.4	39.3
	2%	DB	36.7	43.0	55.4	70.0	79.4	85.8	88.0	85.8	81.4	70.4	56.6	40.5
		MCWB	33.3	38.2	46.5	53.2	60.8	70.2	73.7	71.1	67.8	59.1	49.6	36.5
	5%	DB	33.9	37.1	50.3	63.9	74.8	81.2	85.5	81.4	77.2	64.1	52.1	36.4
		MCWB	32.1	33.9	44.4	49.7	59.2	66.4	71.4	68.2	64.8	55.0	45.6	34.0
	10%	DB	31.8	34.5	45.1	59.2	70.4	77.9	81.2	78.8	73.0	59.4	46.2	34.1
		MCWB	30.4	31.8	39.3	46.9	56.8	64.6	68.8	66.7	63.0	51.2	41.3	32.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.5	43.0	57.1	59.6	69.8	76.3	78.6	75.8	73.6	66.0	55.8	41.5
		MCDB	43.4	48.8	63.2	71.7	80.9	86.8	88.1	85.2	81.8	72.6	61.0	44.2
	2%	WB	33.9	38.6	49.7	54.3	64.4	72.4	75.6	73.0	70.3	60.6	50.7	36.9
		MCDB	35.8	42.5	55.4	65.2	74.4	82.2	84.1	81.6	77.6	67.6	56.2	39.7
	5%	WB	31.8	34.7	44.2	51.1	61.8	69.4	73.1	70.8	67.6	56.3	45.7	34.2
		MCDB	33.0	37.5	50.6	62.2	70.9	77.7	81.8	78.2	74.9	63.7	50.4	36.0
	10%	WB	29.7	32.0	39.5	48.0	58.9	66.9	70.7	68.7	65.1	52.9	42.0	32.2
		MCDB	31.2	33.9	44.7	58.3	68.4	74.8	78.6	75.4	71.9	58.1	47.2	33.5
(35) Mean Daily Temperature Range	5% DB	MDBR	18.3	19.5	18.9	20.6	22.6	21.5	21.2	21.0	21.3	18.4	15.4	15.1
		MCDBR	17.8	19.6	24.4	30.2	30.3	25.7	24.6	24.6	25.8	27.0	24.3	17.6
		MCWBR	15.2	15.5	16.3	17.1	15.5	12.7	12.4	12.7	14.2	16.5	16.4	14.5
	5% WB	MCDBR	16.2	17.7	23.7	27.6	25.2	21.9	22.3	21.2	21.4	23.1	21.8	15.2
		MCWBR	14.5	14.0	16.4	16.7	14.6	12.5	12.7	12.4	13.4	15.5	15.9	12.9
(40) Clear-Sky Solar Irradiance	taub		0.257	0.266	0.312	0.370	0.401	0.387	0.383	0.395	0.366	0.314	0.286	0.261
	taud		2.392	2.364	2.364	2.301	2.263	2.339	2.374	2.352	2.405	2.546	2.508	2.419
	Ebn at Noon		274	293	292	281	275	279	279	271	269	273	261	259
	Edh at Noon		22	28	33	39	42	39	37	36	32	24	20	20
(44) All-Sky Solar Radiation	RadAvg		525	875	1225	1473	1692	1862	1973	1646	1263	784	493	367
	RadStd		58	79	110	140	134	133	70	123	89	104	46	43

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	-1.48	N/A	N/A	-281	-137
Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page